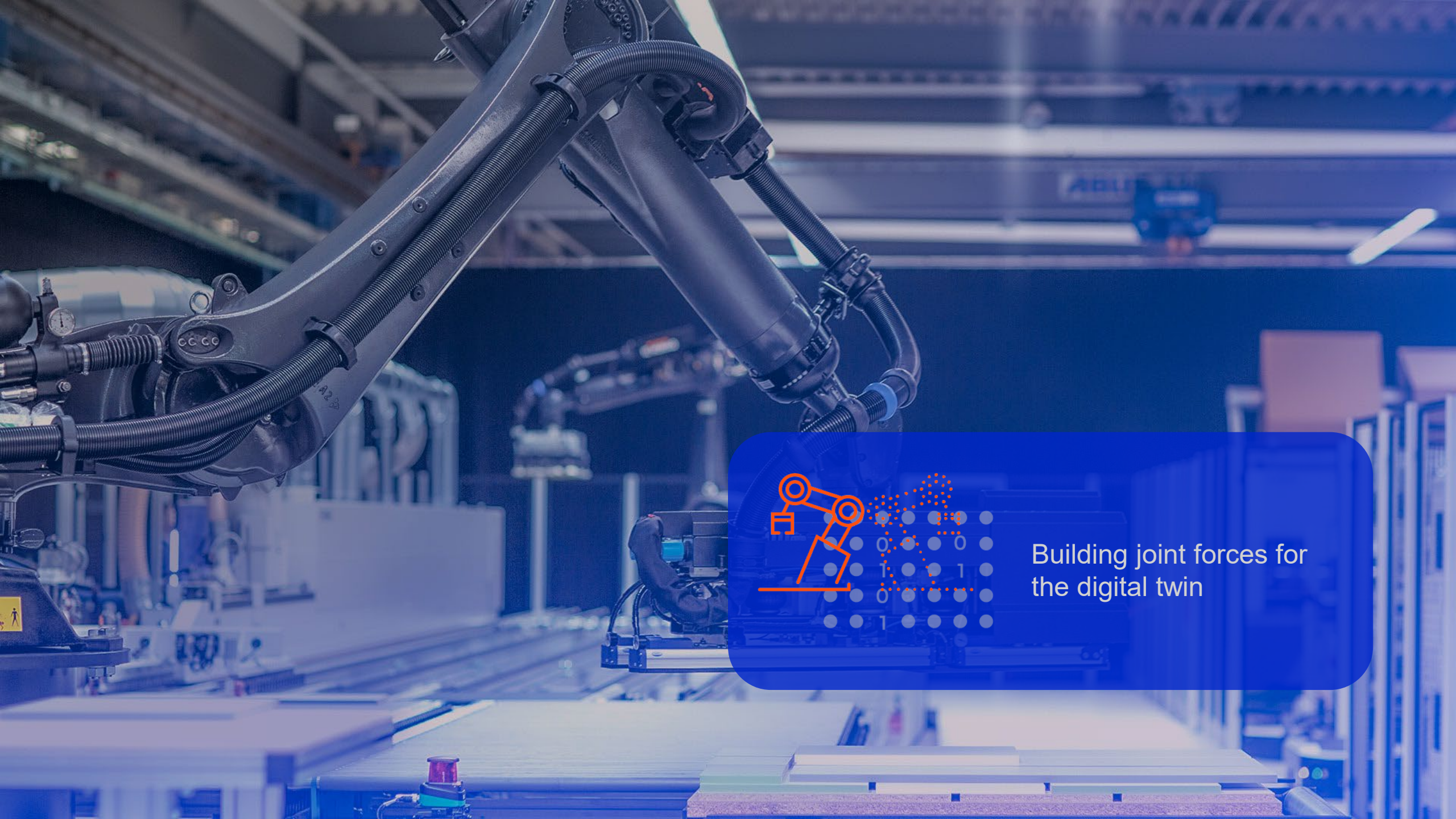
Industrial Digital Twin Association

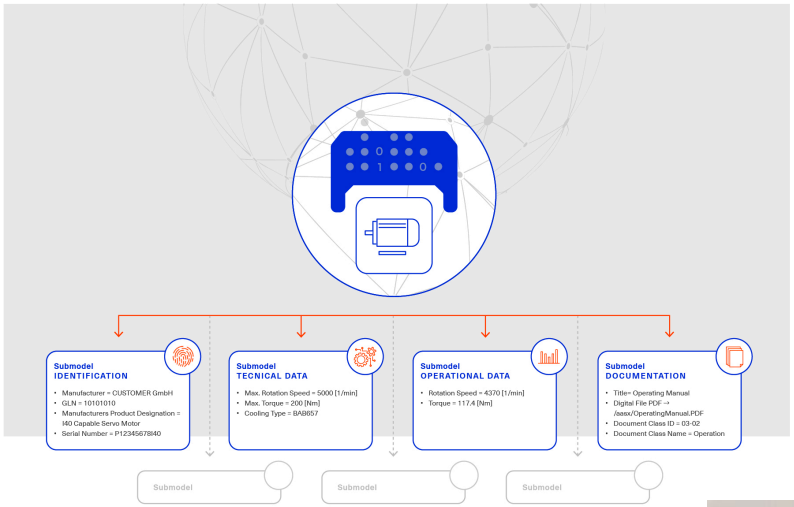
Tutorial

Specification Asset Administration Shell Part 3a Data Specification IEC61360

V3.0 April 2023

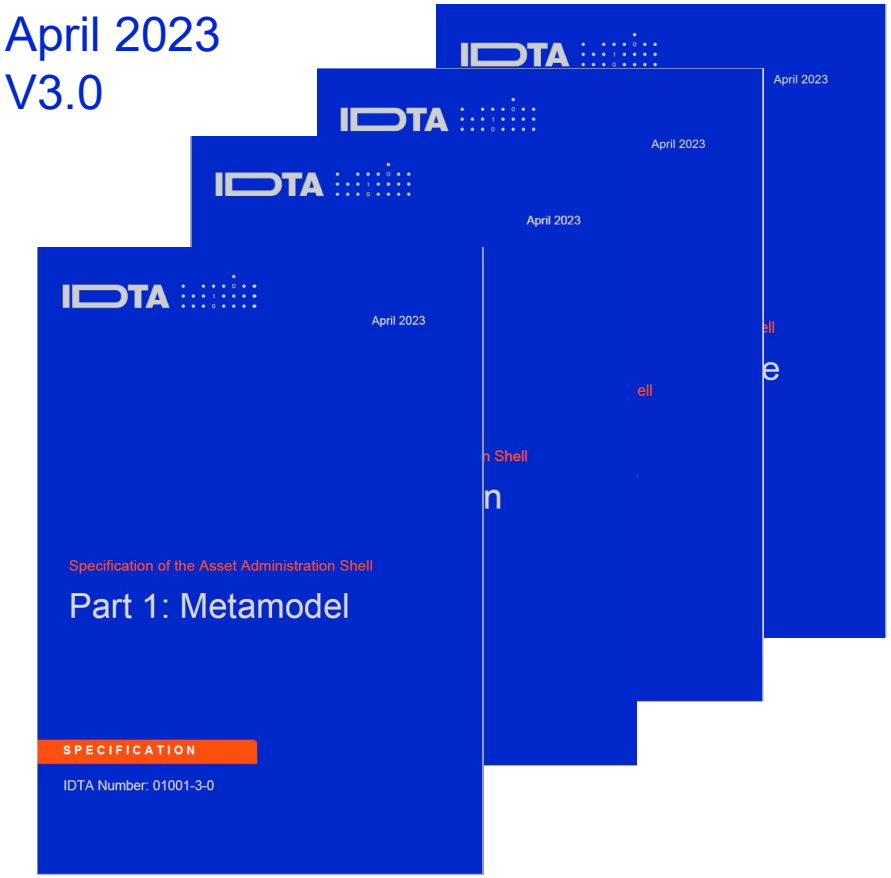

Building joint forces for
the digital twin

From Concept to Specification

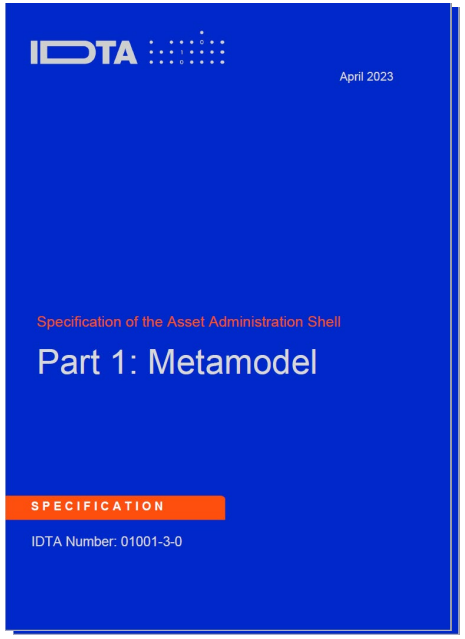


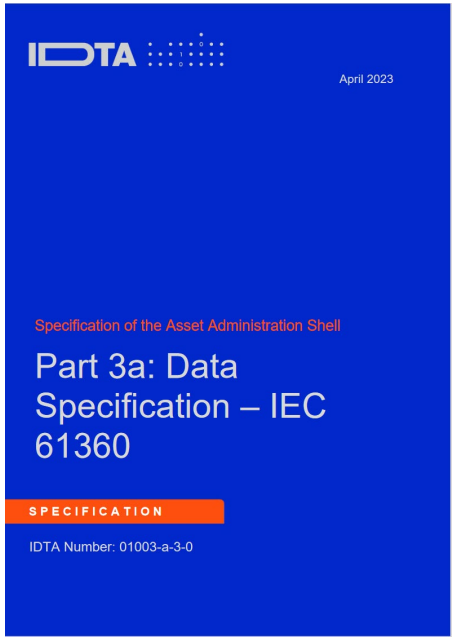
2018
V1.0

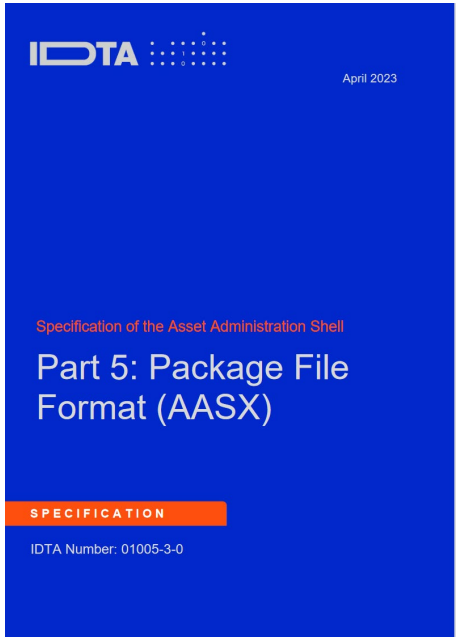
April 2023
V3.0



Specifications







V3.0
April 2023

This Tutorial is about Part 3a

IEC 63278

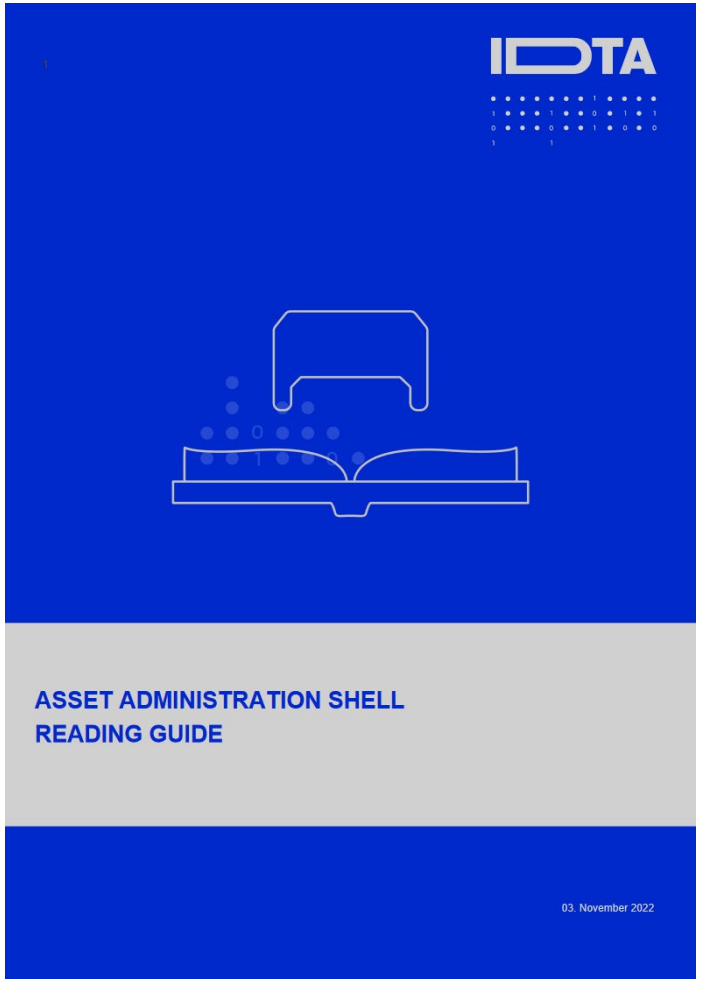
For whom is this tutorial?

- You should have an idea of the benefits of introducting digital twins to your domain
- You should know Tutorial Part 1 Metamodel
- You should have basic knowledge in UML modeling
- You are an architect and want to learn more about the underlying information model of the Asset Administration Shell
- You are a developer and want to upgrade to the new version or start your first implementation

For whom is this tutorial?

- You feel dissappointed and do not know how to start?
  Have a look at the Asset Administration Shell Reading Guide!
  It is updated on a regular basis.

- **Where to start:** If you have never heard of the AAS
- **For the generally interested reader:** If you want to learn more about the subject
- **For decision makers:** If you are interested in the business side of I4.0
- **For software developers and architects:** If you want to know how to create software for the AAS
- **For users of the AAS and domain experts:** If you are interested in using the AAS for specific tasks
- **Security and AI:** If you want to deep dive into these special topics.



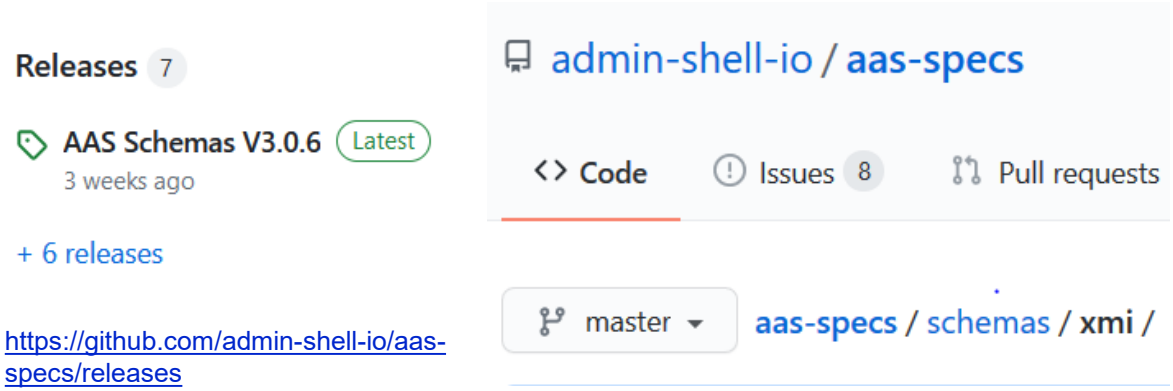
Get started

Download Specification

https://industrialdigitaltwin.org/content-hub/

Import XMI to your UML tooling

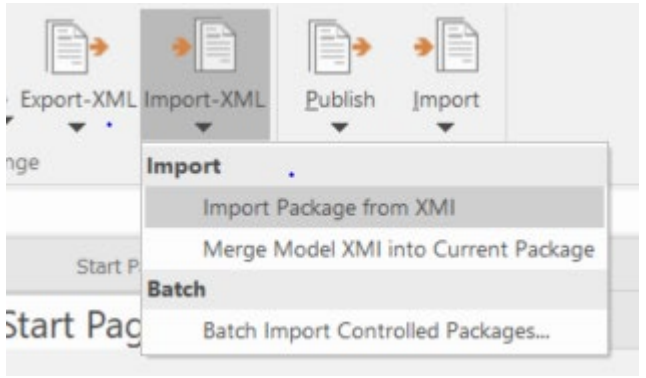
1.



https://github.com/admin-shell-io/aas-specs/releases

https://github.com/admin-shell-io/aas-specs/tree/master/schemas/xmi

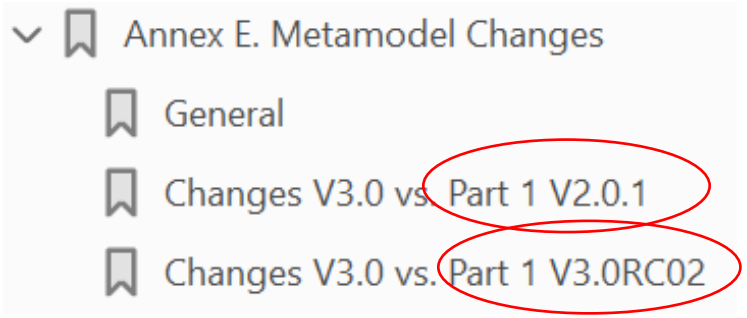
2.



1. Fetch release of AAS you are interested in
2. Import xmi file into UML tool (best with Enterprise/Architect)

Note: The data specifications are embedded into the Part 1 schemas

Metamodel Changes Part 3a



Note for Experts: <Notes for tutorial listeners who have knowledge of previous versions of the specification (V2.0 or Release Candidates of V3.0).

If you do not know previous versions you can ignore these notes.>

14 - Birgit Boss: Details of the Asset Administration Shell V3.0 (Release...

V3.0RC02

https://www.youtube.com/watch?v=QR-nOl6cuOI

UML Legend

1. Get (re-)familiar with general UML modeling rules
2. Get familiar with specific graphical representation of UML in the specification (partly tool specific)

Annex D. Legend for UML Modelling

OMG UML General

This annex explains the UML elements used in this specification. For more information, please refer to the comprehensive literature available for UML. The formal specification can be found in [12].

Figure 20 shows a class with name "Class1" and an attribute with name "attr" of type *Class2*. Attributes are owned by the class. Some of these attributes may represents the end of binary associations, see also Figure 21. In this case, the instance of *Class2* is navigable via the instance of the owning class *Class1*.[14]

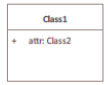


Figure 20 Class

Figure 21 shows that *Class4* inherits all member elements from *Class3*. Or in other word, *Class3* is a generalization of *Class4, Class4* is a specialization of *Class3*. This means that each instance of *Class4* is also an instance of *Class3*. An instance of *Class4* has the attributes *attr1* and *attr2*, whereas instances of *Class3* only have the attribute *attr1*.

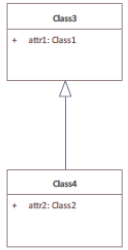
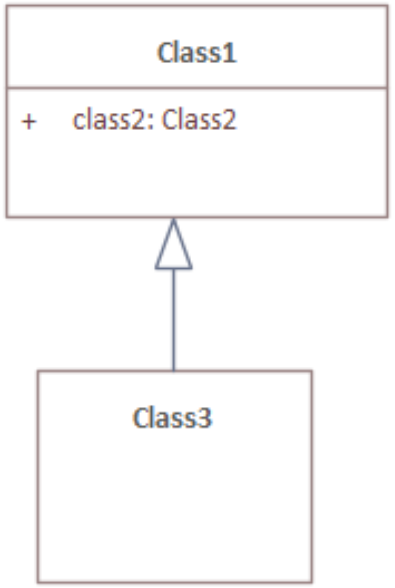


Figure 21 Inheritance/Generalization

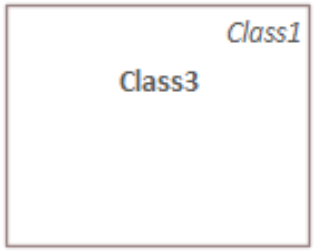
Figure 22defines the required and allowed multiplicity/cardinality within an association between instances of

UML Generalization/Inheritance

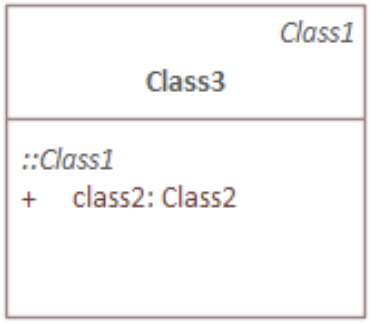
A)



B)



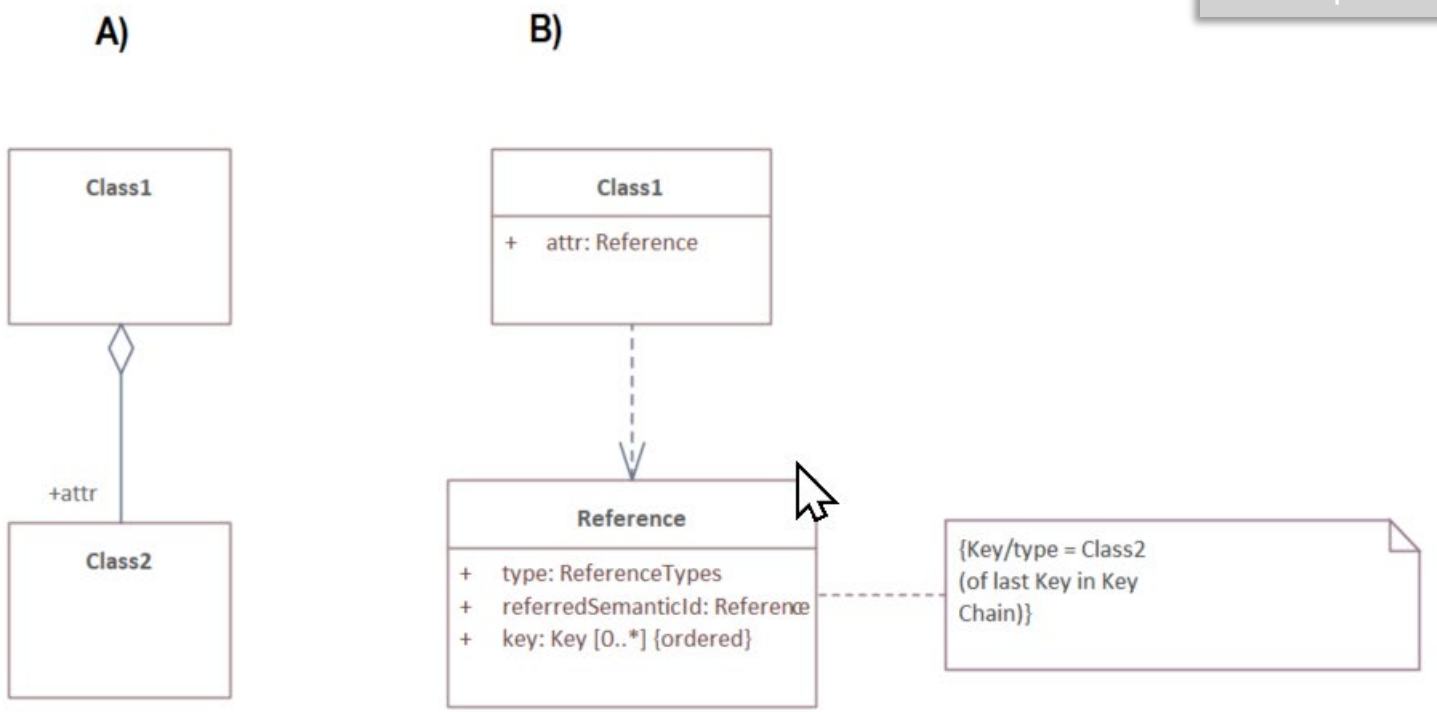
C)



Hint: Graphical representation tool specific

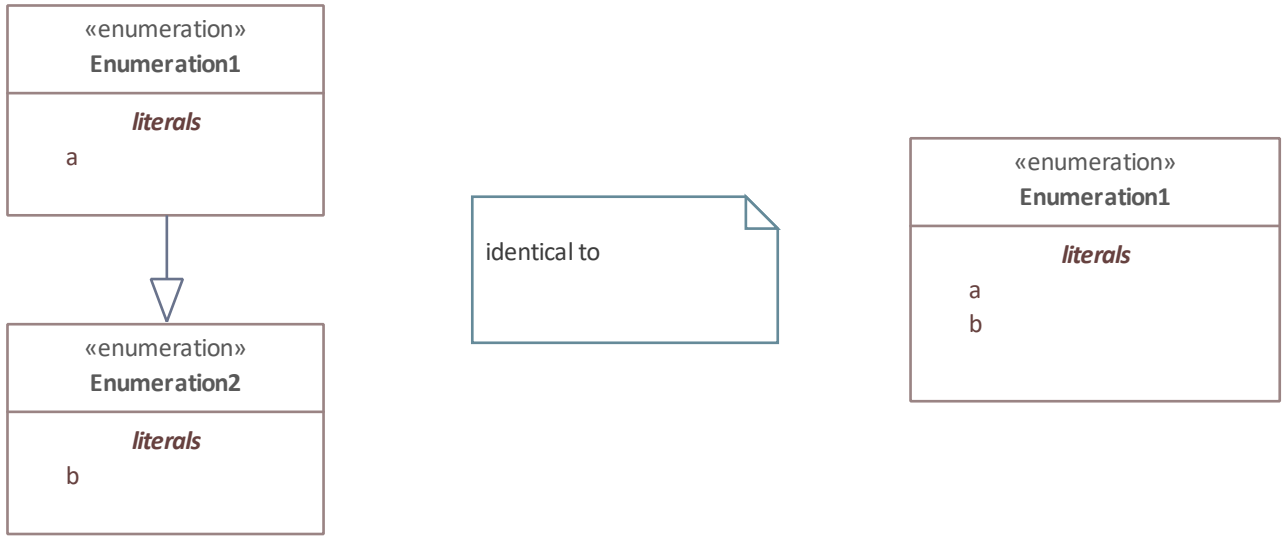
UML Shared Aggregation

Note: Referencing of Referables is an important concept to understand when implementing the AAS



Note for experts: In previous versions a notation of class attributes with reference (*) was used additionally to the notation with the association with the diamond.

Enumerations



Note 1: Inheritance between enumerations is not widely used. It is only used for graphical illustration of relationships between enumerations

Templates

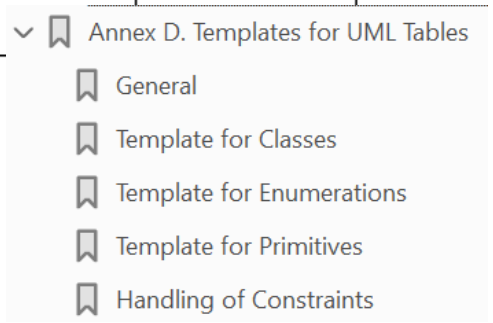
Template for Classes

Template for Classes:

Class:	<Class Name> [<<abstract>>] ["<<Experimental>>"] ["<<Deprecated>>"] ["<<Template>>"]		
Explanation:	<Explanatory text>		
Inherits from:	{<Class Name> ";" }+ | "-"		
Attribute	**Explanation**	**Type**	**Card.**
<attribute or association name> ["<<ordered>>"] ["<<Experimental>>"] ["<<Deprecated>>"]	<Explanatory text>	<Type>	<Card>

Note for experts: ModelReference<SubmodelElement> is equal to former notation SubmodelElement*

Note for experts: no kind column any longer, instead different notation for Type



Example for Class Specification

5.3.7.12 Property Attributes

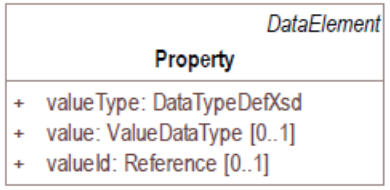
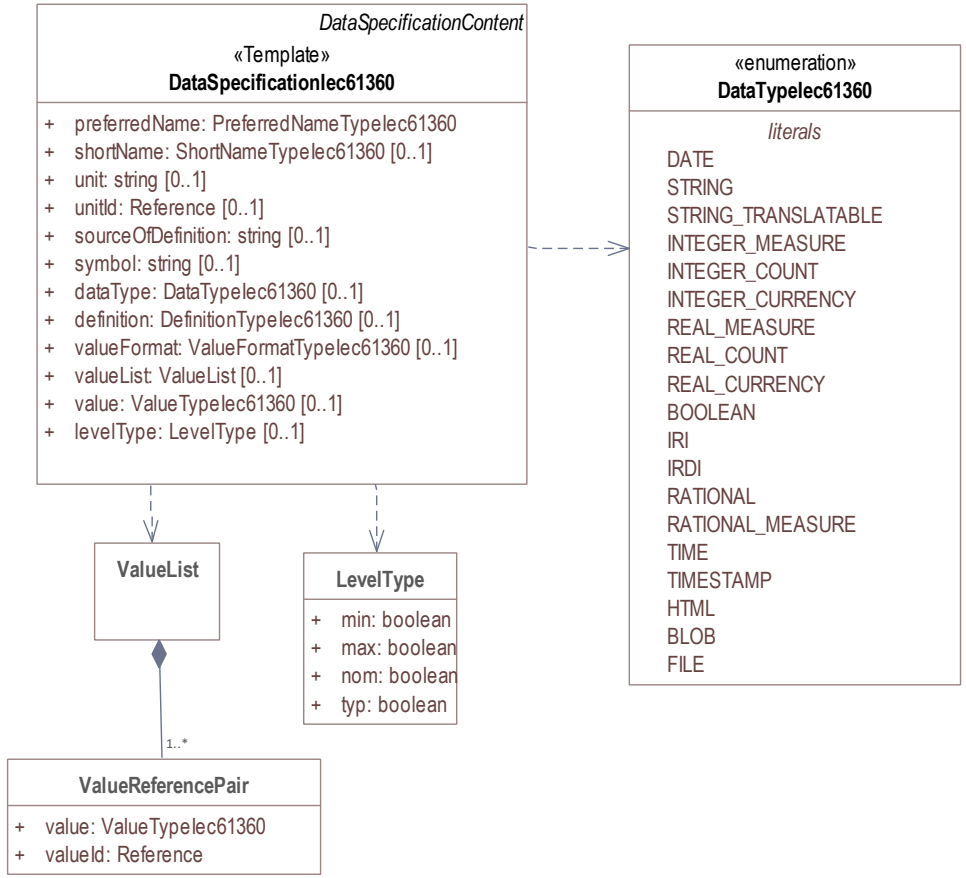


Figure 39 Metamodel of Properties

Class:	Property		
Explanation:	A property is a data element that has a single value. Constraint AASd-007: If both the *Property/value* and the *Property/valueId* are present, the value of *Property/value* needs to be identical to the value of the referenced coded value in *Property/valueId*.		
Inherits from:	DataElement		
Attribute	**Explanation**	**Type**	**Card.**
valueType	Data type of the value attribute	DataTypeDefXsd	1
value	The value of the property instance	ValueDataType	0..1
valueId	Reference to the global unique ID of a coded value Note: it is recommended to use an external reference.	Reference	0..1

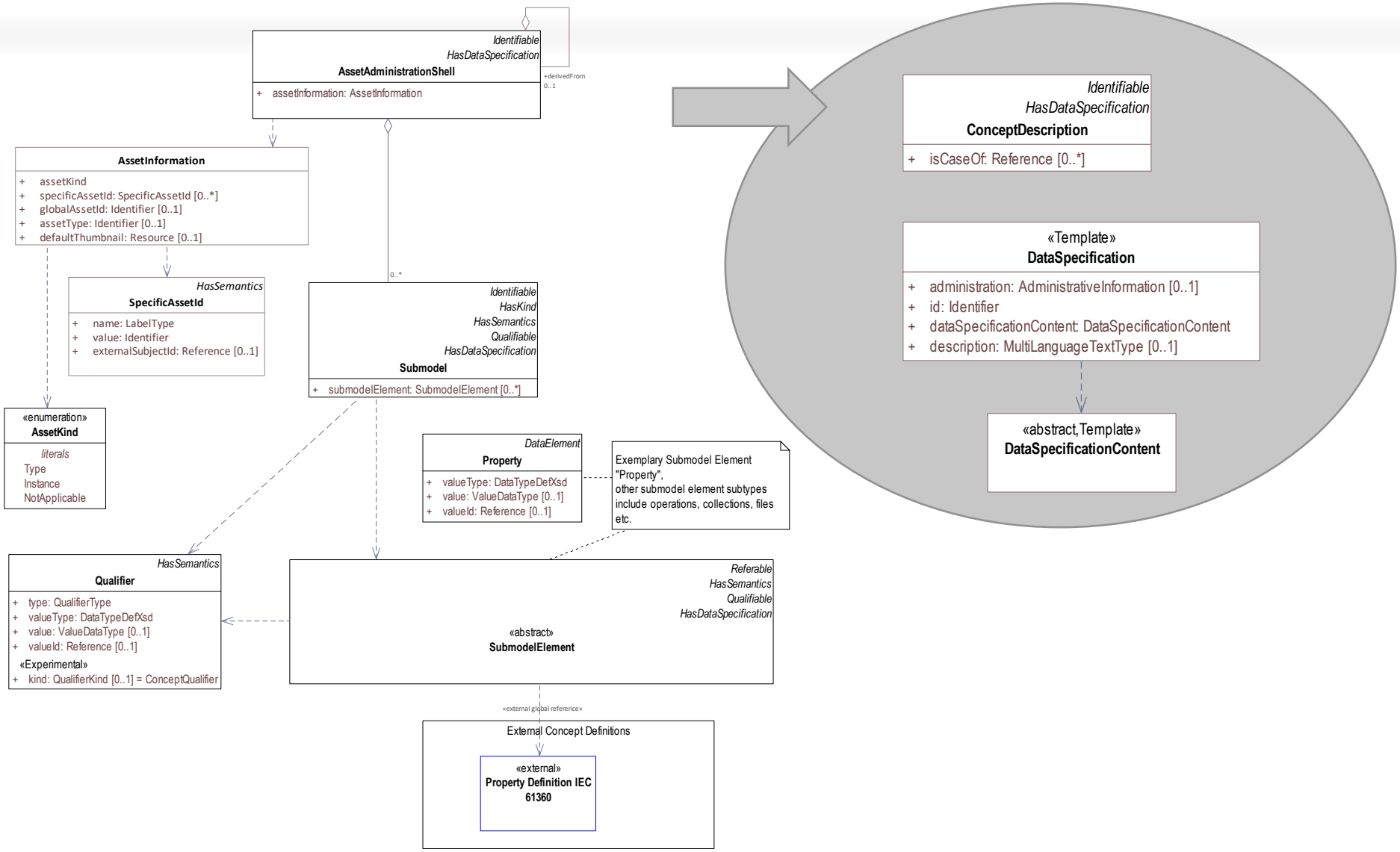
Get warm – Prerequisites from PART 1 Metamodel

Overview – the goal is to be able to read this diagram

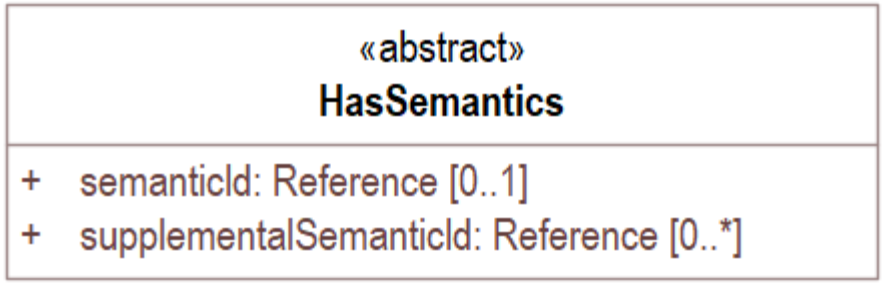


Data Specification for Concept Descriptions (see Part 1)

Part 1 – Data Specifications & Concept Descriptions



Part 1 Common - HasSemantics



The semanticId is the identifier of the semantic definition of the element.
Supplemental semantic IDs can be added.
The semanticId can be an External Reference or a Model Reference to a **Concept Description**.

Part 1 Common - HasDataSpecification

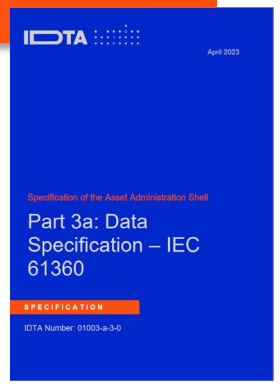
«abstract» **HasDataSpecification**
+ dataSpecification: Reference [0..*]

Allows to define standardized templates for data specification

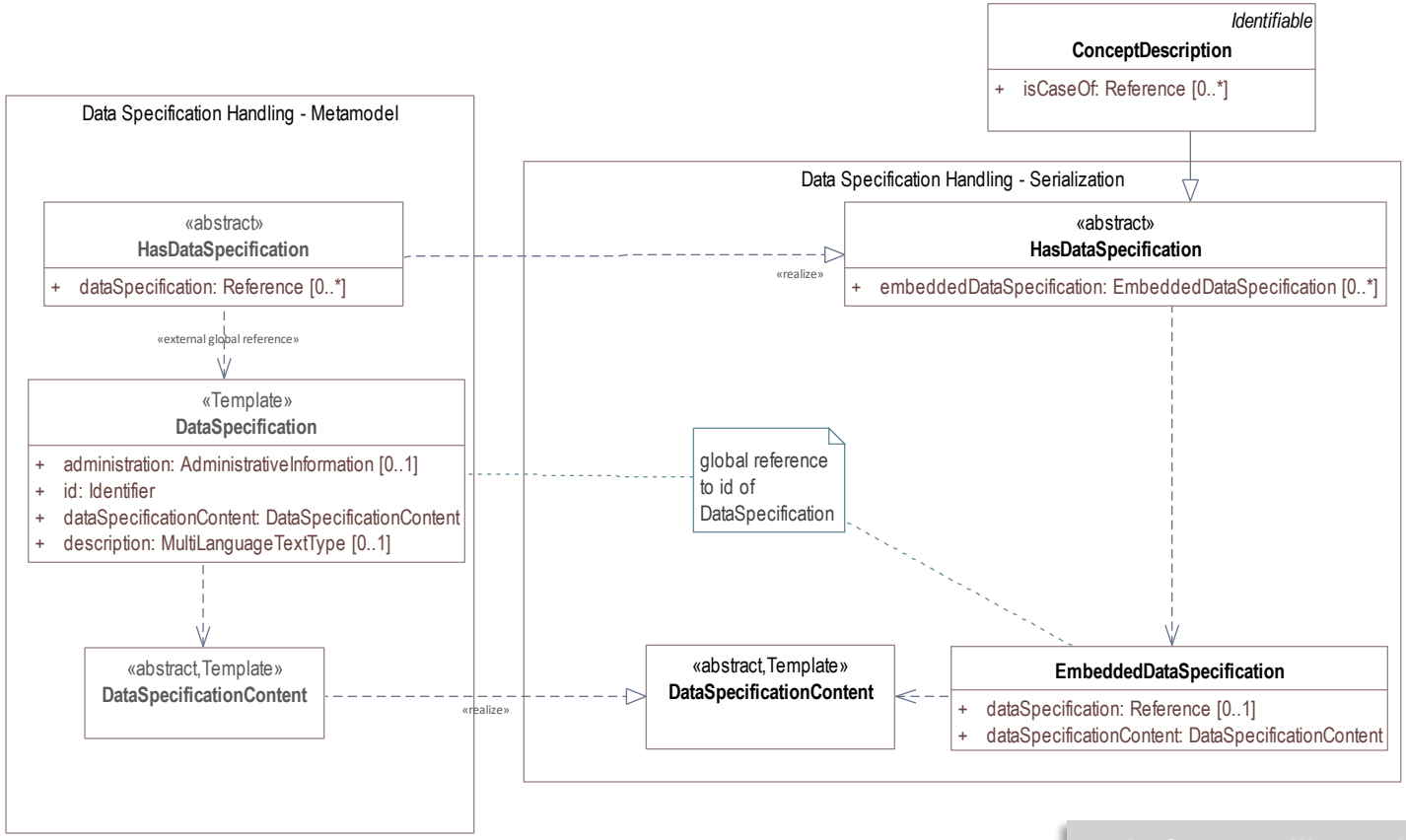
Data specification reference shall be globally unique and identifies which data specifications are used for an object

Attributes defined in template are added to the object

Note for Experts: Data Specifications are not part of Part 1 any longer: They are part of data specifications series Part 3



Part 1 - Embedded Data Specifications



Property	0173-1#02-BAA120#008 Max. rotation speed
Data type	INTEGER_MEASURE
Unit of measure	1/min
Definition	Greatest possible rotation speed with wich the motor or feeding unit may be operated

DataSpecificationContent

«Template»

DataSpecificationIec61360

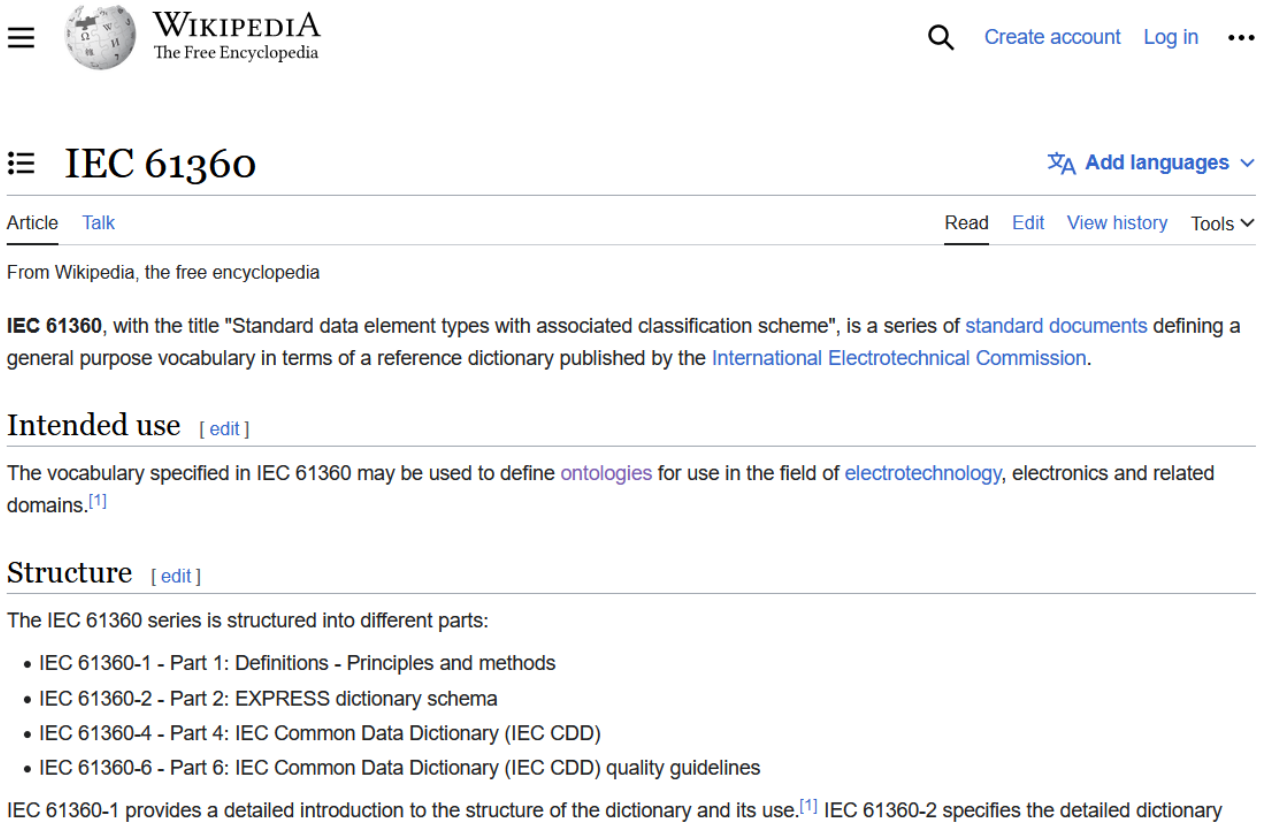
+ preferredName: PreferredNameTypeIec61360
+ shortName: ShortNameTypeIec61360 [0..1]
+ unit: string [0..1]
+ unitId: Reference [0..1]
+ sourceOfDefinition: string [0..1]
+ symbol: string [0..1]
+ dataType: DataTypeIec61360 [0..1]
+ definition: DefinitionTypeIec61360 [0..1]
+ valueFormat: ValueFormatTypeIec61360 [0..1]
+ valueList: ValueList [0..1]
+ value: ValueTypeIec61360 [0..1]
+ levelType: LevelType [0..1]

In formats like xml, JSON, rdf the embedded data specification approach is implemented

Data Specifications in Part 3 – data specification template IEC 61360 defined in Part 3a

i700E70
231231 Serial number
0.02 Inverter current
Serial number
Inverter current
IEC 61360

IEC 61360



Wikipedia
The Free Encyclopedia

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IEC 61360

Add languages

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From Wikipedia, the free encyclopedia

IEC 61360, with the title "Standard data element types with associated classification scheme", is a series of standard documents defining a general purpose vocabulary in terms of a reference dictionary published by the International Electrotechnical Commission.

Intended use [edit]

The vocabulary specified in IEC 61360 may be used to define ontologies for use in the field of electrotechnology, electronics and related domains.[1]

Structure [edit]

The IEC 61360 series is structured into different parts:

- IEC 61360-1 - Part 1: Definitions - Principles and methods
- IEC 61360-2 - Part 2: EXPRESS dictionary schema
- IEC 61360-4 - Part 4: IEC Common Data Dictionary (IEC CDD)
- IEC 61360-6 - Part 6: IEC Common Data Dictionary (IEC CDD) quality guidelines

IEC 61360-1 provides a detailed introduction to the structure of the dictionary and its use.[1] IEC 61360-2 specifies the detailed dictionary data model and IEC 61360-6 stipulates quality criteria for the content of the dictionary.[1] The data model defined in IEC 61360-2 is also published in ISO 13584-42.[2]

The IEC provides a technical dictionary for the use in the electro-technical and electronic domain which is published as IEC 61360-4. This dictionary is called IEC Common Data Dictionary (IEC CDD) and can be accessed as a web page (https://cdd.iec.ch).

See also [edit]

IEC 61360 also defines the base for other product taxonomies like eCl@ss.

Industrie 4.0 uses product property description based on IEC 61360.[3]

- "IEC 61360" at International Electrotechnical Commission

Note: Always be aware of usage policies and licensing before using a predefined/standardized concept description!

Status 2023-09

IEC 61360 Property

Preferred name	Max. rotation speed
IRDI	0173-1#02-BAA120#008
Definition	Greatest permissible rotation speed with which the motor or feeding unit may be operated
Short name of unit	1/min
Quantity	frequency
Type of Property	Non-dependent
Valency type	Multivalent
Definition class	ECLASS (0173-1#01-RAA001#001)
Property data type	Integer (measure)
Class type code	F03 - frequency, rotational frequency
Allow negative values	false
Property Original Identifier	BAA120001

Example measurable Property from ECLASS

IEC 61360 Value List

Property	02-BAE122 Cooling type
short name	-
Format	STRING
Definition:	Summary of various types of cooling, for use as search criteria that limit a selection
Values:	

0173-1#07-BAB649#001 - Air-air heat exchanger
0173-1#07-BAB650#001 - Air-water heat exchanger
0173-1#07-BAB592#001 - alien
0173-1#07-BAB611#001 - closed, external air-cooling
0173-1#07-BAB610#001 - closed, internal air-cooling
0173-1#07-BAB591#003 - free cooling
0173-1#07-BAB702#003 - Heat exchanger against other cooling medium
0173-1#07-BAB657#003 - open circuit, external cooling
0173-1#07-BAB656#003 - open circuit, internal cooling
0173-1#07-BAB535#003 - other form of cooling with primary air coolant
0173-1#07-BAB536#003 - other primary non-air coolant
0173-1#07-BAB674#003 - self

Example Value List from ECLASS

IEC 61360 Value

Property	02-BAE122 Cooling type
short name	-
Format	STRING
Definition:	Summary of various types of cooling, for use as search criteria that limit a selection
Values:	

0173-1#07-BAB649#001 - Air-air heat exchanger
0173-1#07-BAB650#001 - Air-water heat exchanger
0173-1#07-BAB592#001 - alien
0173-1#07-BAB611#001 - closed, external air-cooling
0173-1#07-BAB610#001 - closed, internal air-cooling
0173-1#07-BAB591#003 - free cooling
0173-1#07-BAB702#003 - Heat exchanger against other cooling medium
0173-1#07-BAB657#003 - open circuit, external cooling
0173-1#07-BAB656#003 - open circuit, internal cooling
0173-1#07-BAB535#003 - other form of cooling with primary air coolant
0173-1#07-BAB536#003 - other primary non-air coolant
0173-1#07-BAB674#003 - self

Value	0173-1#07-BAB657#003
Classification	open circuit, external cooling
short name	
Definition:	

Change Text | Replace | Delete | Copy

General | Admin | Attribute | CR | History | Release

Value	BAB657
IRDI	0173-1#07-BAB657#003
eCl@ss v5 ID	BAB657001
Preferred Name	open circuit, external cooling
Short Name	
Definition	
Source of Definition	
Note to Definition	
Data Type	String
Value specification	Coded values
Exception	No

Example Value from ECLASS and ECLASS Advanced

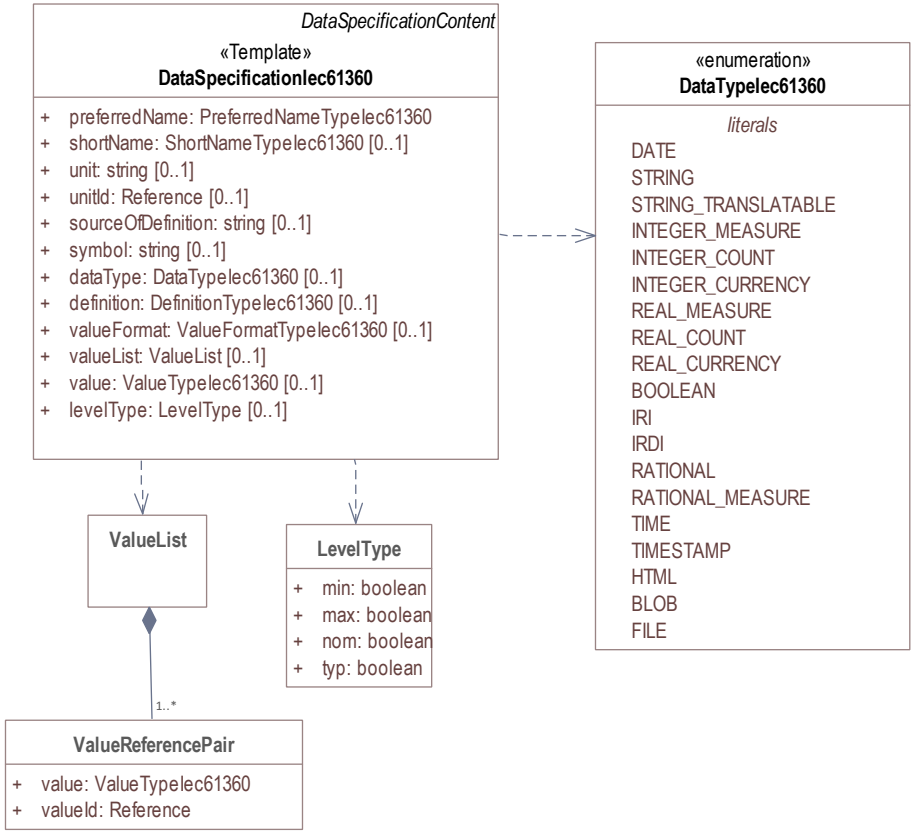
IEC 61360 Level Type

Code:	0112/2///61360_4#AAE022
Version:	001
Revision:	05
IRDI:	0112/2///61360_4#AAE022#001
Preferred name:	outside diameter
Synonymous name:	
Symbol:	d_{out}
Synonymous symbol:	
Short name:	d_out
Definition:	value as specified by level (miNomax) of the outside diameter of a component with a body of circular cross-section
Note:	
Remark:	
Primary unit:	m
Alternative units:	
Level:	miNoMax
Data type:	LEVEL(MIN,MAX,NOM) OF REAL_MEASURE_TYPE
Format:	NR3..3.3ES2
Property constraint:	
Definition source:	
Value source:	
Property data element type:	NON_DEPENDENT_P_DET
Drawing:	
Formula:	
Value list code:	
Value list:	
DET class:	T03
Applicable classes:	0112/2///61360_4#AAA001 - component
Definition class:	0112/2///61360_4#AAA001
Code for unit:	0112/2///62720#UAA726 - metre
Codes for alternative units:	
Code for unit list:	

Example Property with Level Type MIN, MAX and NOM from IEC CDD

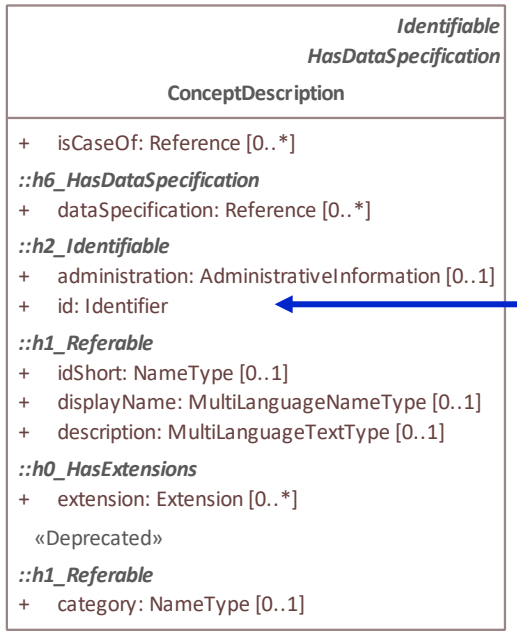
i700E70
231231 Serial number
0.02 Inverter current
Serial number
Inverter current
IEC 61360

Data Specification IEC 61360



Proprety	0173-1#02-BAA120#008 Max. rotation speed
Data type	INTEGER_MEASURE
Unit of measure	1/min
Definition	Greatest possible rotation speed with wich the motor or feeding unit may be operated

Concept Description



Unique ID of property, value list or value defined via the data specification IEC62360

Note 1: IEC 61360 also requires a globally unique identifier for a concept description. This ID is not part of the data specification template. Instead, the *ConceptDescription/id* as inherited via *Identifiable* is used. The same applies to administrative information like the version and revision.

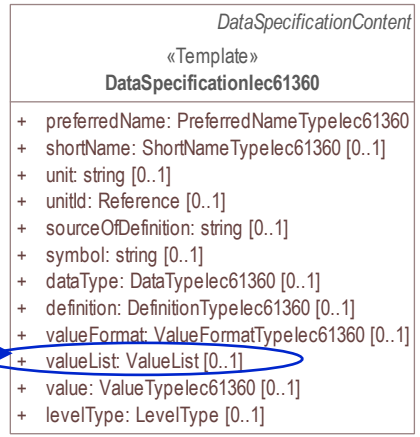
Note 2: *ConceptDescription/idShort* and *DataSpecificationIec61360/shortName* are very similar. However, in this case, *shortName* is explicitly added to the data specification.

Note 3: the same applies to *ConceptDescription/displayName* and *DataSpecificationIec61360/preferredName*.

Note 4: the same applies to *ConceptDescription/description* and *DataSpecificationIec61360/definition*.

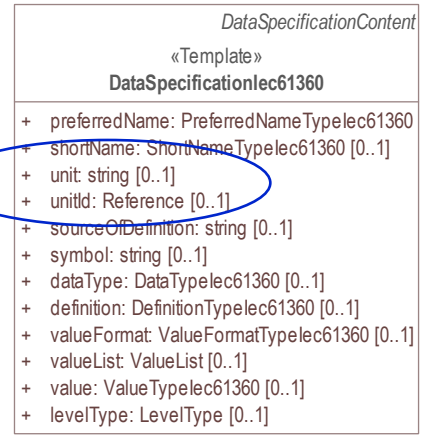
Data Specification IEC 61360

For Value Lists:

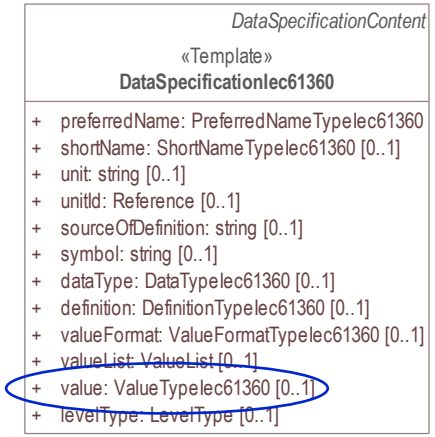


For measurable properties

(dataType = *_MEASURE)

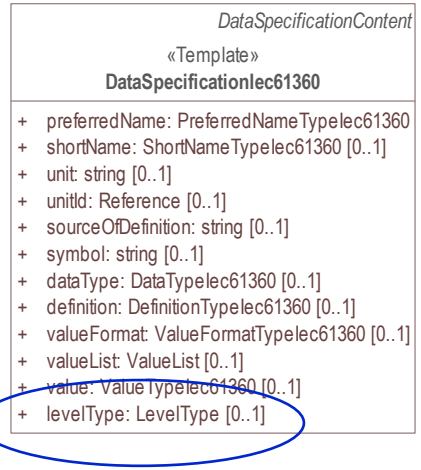


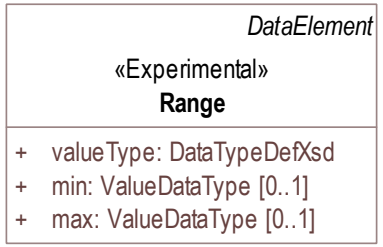
For values (within value lists)



IEC 61360 Level Types and Ranges

For properties with Level Type

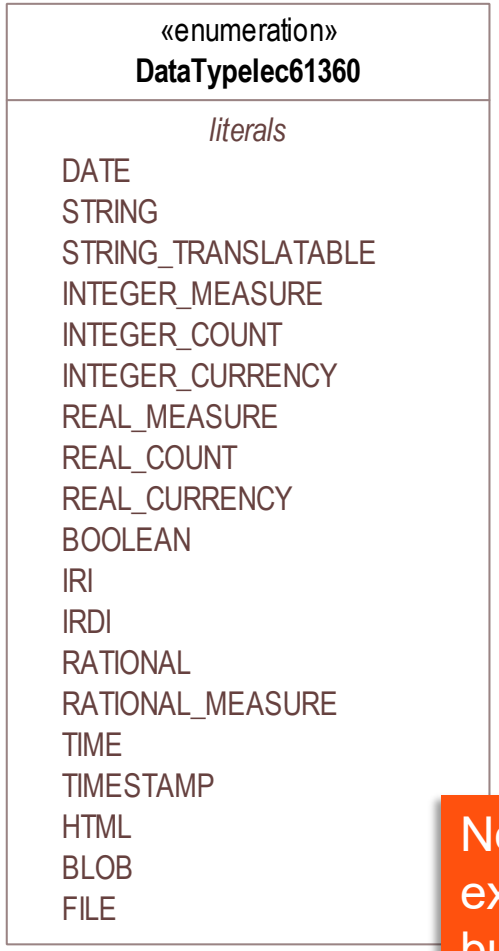




Use level type MIN and MAX for submodel element „Range“ (see Part 1)

It is not recommended to use standardized properties with level types except those defining ranges.

Data Types IEC 61360

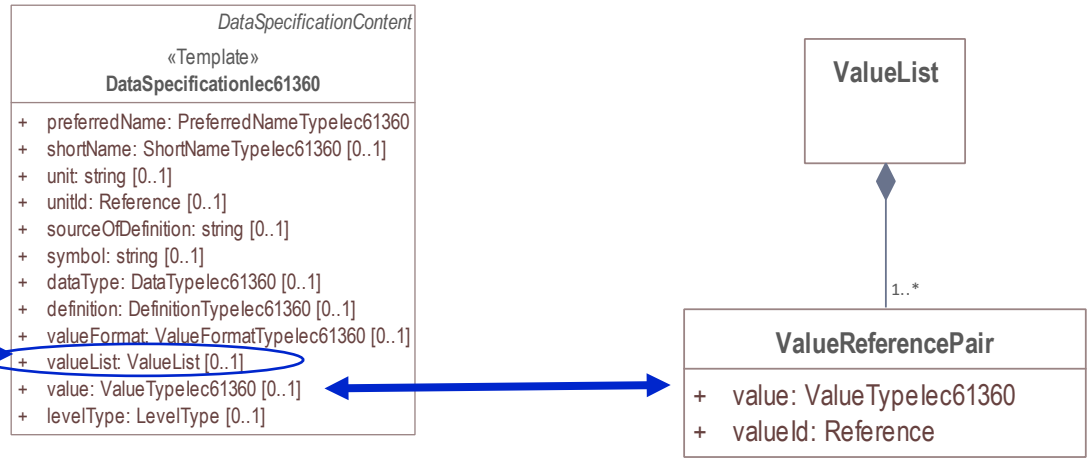


Data Type IEC 61360	xsd Value Type[3]	Example Values IEC 61360[4]
DATE	xs:date	1979-01-15
STRING	xs:string	"DN 700" "10 Mbps"
STRING_TRANSLATABLE	*Mapped to MultiLanguageProperty, i.e. type MultiLanguageText* Note: for details, please see Part 1 of the document series "Details of the Asset Administration Shell".	
INTEGER_MEASURE	xs:integer	1 10 111
INTEGER_COUNT	xs:integer	1 10 111
INTEGER_CURRENCY	xs:integer	1 10 111
REAL_MEASURE	xs:double or xs:float (depending on needed	1.5 102.35

Note: Data Type FILE is not explicitly supported by IEC61360 but used for data elements File

Value List

For Value Lists:

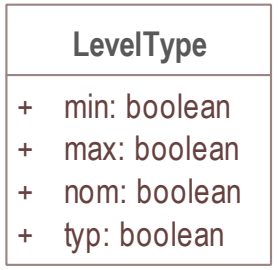


The valueId references a value

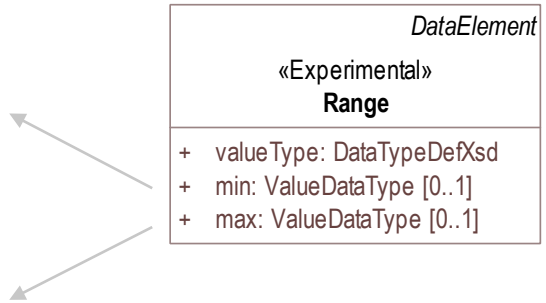
Property	02-BAE122 Cooling type
short name	-
Format	STRING
Definition:	Summary of various types of cooling, for use as search criteria that limit a selection
Values:	

0173-1#07-BAB649#001 - Air-air heat exchanger
0173-1#07-BAB650#001 - Air-water heat exchanger
0173-1#07-BAB592#001 - alien
0173-1#07-BAB611#001 - closed, external air-cooling
0173-1#07-BAB610#001 - closed, internal air-cooling
0173-1#07-BAB591#003 - free cooling
0173-1#07-BAB702#003 - Heat exchanger against other cooling medium
0173-1#07-BAB657#003 - open circuit, external cooling
0173-1#07-BAB656#003 - open circuit, internal cooling
0173-1#07-BAB535#003 - other form of cooling with primary air coolant
0173-1#07-BAB536#003 - other primary non-air coolant
0173-1#07-BAB674#003 - self

Level Type

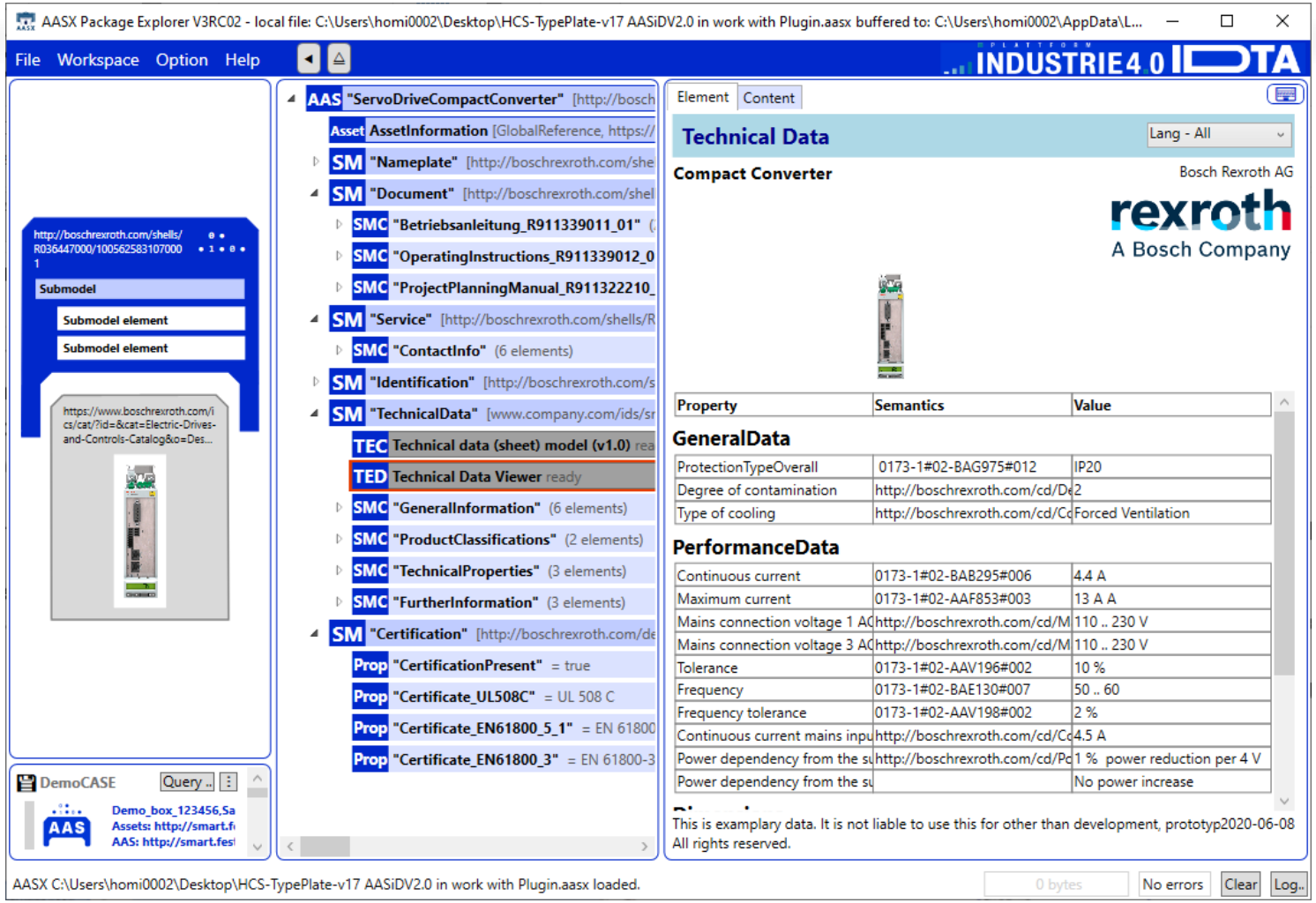


Class:	LevelType		
Explanation:	Value represented by up to four variants of a numeric value in a specific role: MIN, NOM, TYP, and MAX. True means that the value is available, false means the value is not available. Note: for details, please refer to [IEC61360-1], LEVEL_TYPE EXAMPLE from [IEC61360-1]: in case of a property which is of the LEVEL_TYPE min/max – Note: for details, please refer to [IEC61360-1], LEVEL_TYPE		
Inherits from:	DataSpecificationContent		
Attribute	**Explanation**	**Type**	**Card.**
min	Minimum of the value	boolean	1
nom	Nominal value (value as designated)	boolean	1
typ	Value as typically present	boolean	1
max	Maximum of the value	boolean	1



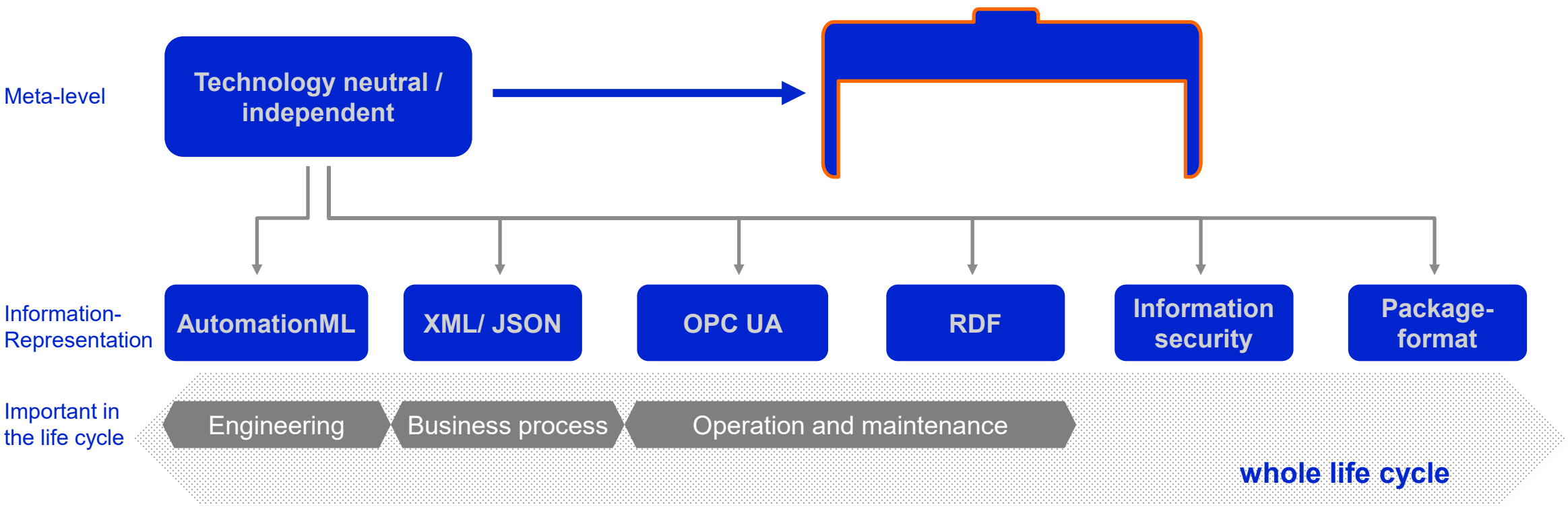
Create your first digital twin

AASX Package Explorer

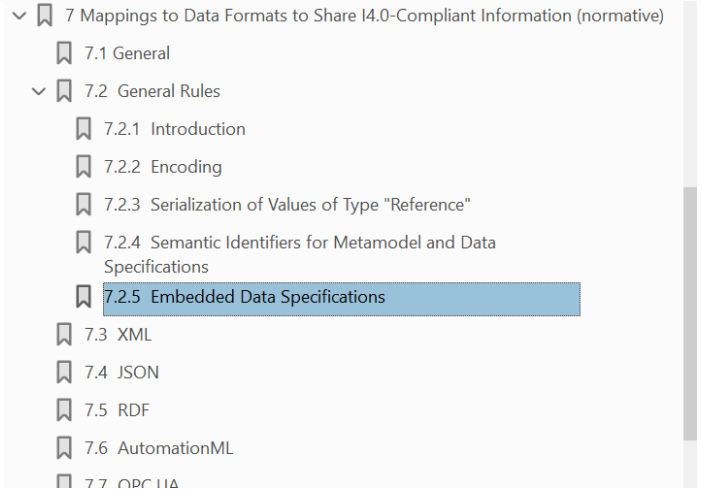


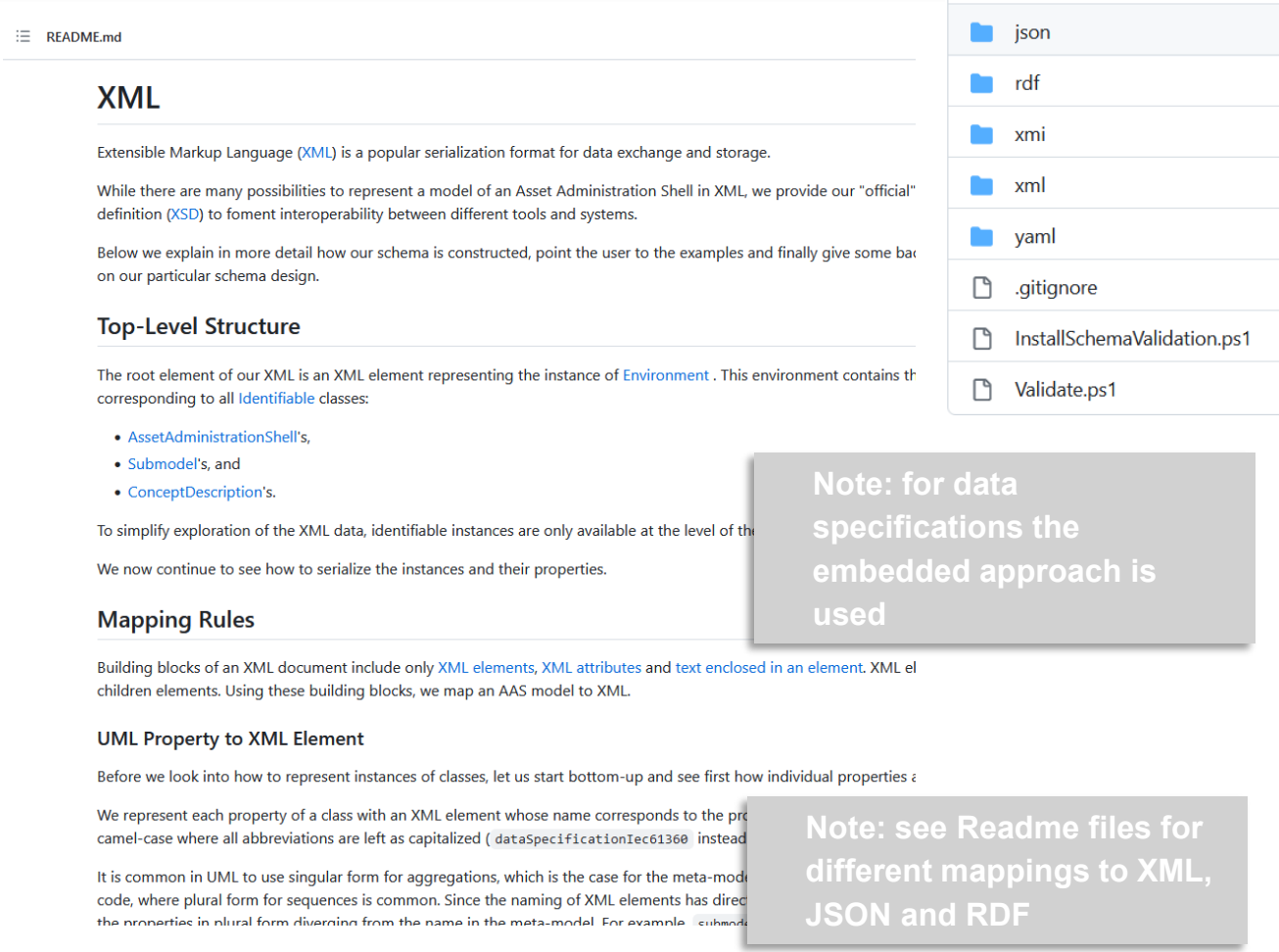
https://github.com/admin-shell-io/aasx-package-explorer

Life Cycle Phases and Formats



Serializations/Formats





https://github.com/admin-shell-io/aas-specs/tree/master/schemas

Note for Experts: Mapping Rules and Schema for xml, JSON and rdf as well as examples not part of specification any longer ➔ now part of open source project admin-shell-io/aas-specs

Note for Experts: Formats like OPC UA or AutomationML are maintained in OPC Foundation and Automation e.V.

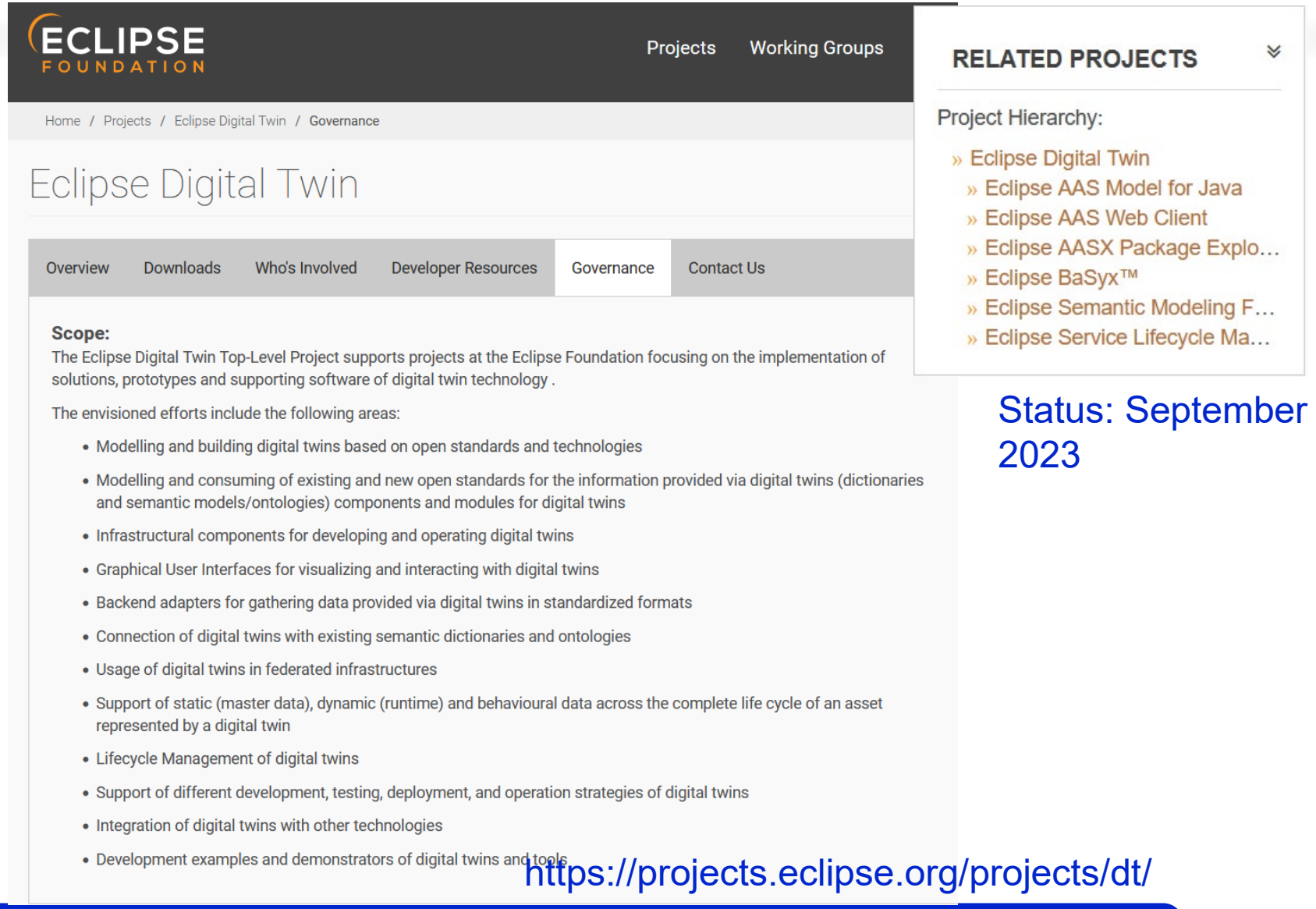
Note: for data specifications the embedded approach is used

Note: see Readme files for different mappings to XML, JSON and RDF

Open Source Support

https://github.com/orgs/admin-shell-io/

Note: specifications maintained in admin-shell-io

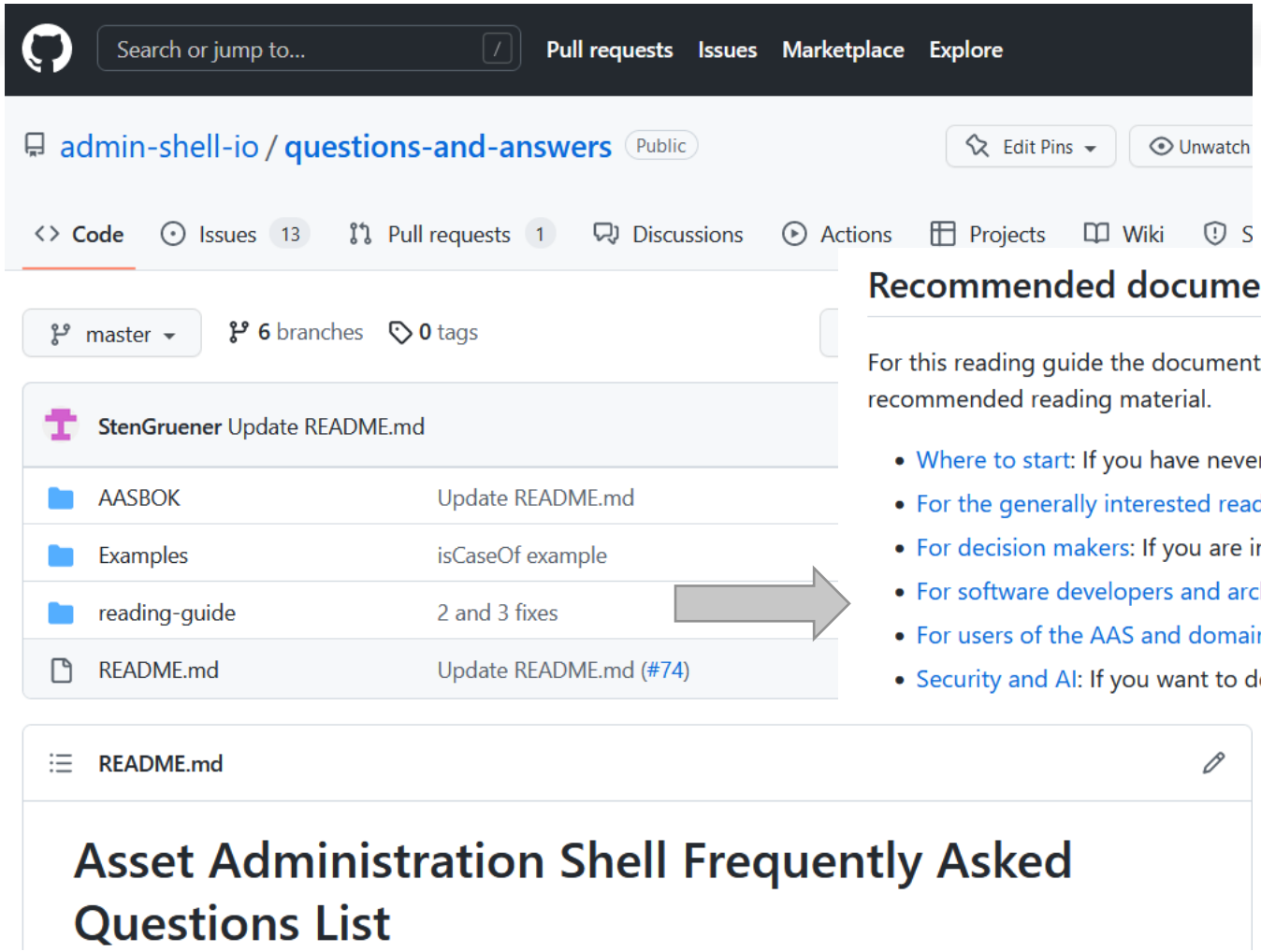


Status: September 2023

https://projects.eclipse.org/projects/dt/

Still Questions?

Questions and Answers



Recommended documents

For this reading guide the documents have been sorted by interest groups rather than topics. In some cases, only specific pages or sections ar recommended reading material.

- Where to start: If you have never heard of the AAS
- For the generally interested reader: If you want to learn more about the subject
- For decision makers: If you are interested in the business side of I4.0
- For software developers and architects: If you want to know how to create software for the AAS
- For users of the AAS and domain experts: If you are interested in using the AAS for specific tasks
- Security and AI: If you want to deep dive into these special topics.

https://github.com/admin-shell-io/questions-and-answers

AAS Academy

www.u4i.io/IDTA

Industrial Digital Twin Association

Let´s go!

Dr. Birgit Boss

Robert Bosch GmbH, Bosch Connected Industry

- Board member of the Industrial Digital Twin Association (IDTA)
- Chair of the Working Group “Open Technology” and its Working Stream “Specifications of the Asset Administration Shell”
- Chair of the Working Group “Semantic Layer including Digital Twins” of Catena-X
- PMC member of the Eclipse Digital Twin Top Level Project

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Industrial Digital Twin Association

www.industrialdigitaltwin.org